

CORE PACK

```
\[
\boxed{
\begin{aligned}
&\text{Helices / Filaments:} \\
&X(s) = \sum_{k=1}^N R_k \mathcal{H}_\ast(k,s), \quad
\mathcal{H}_\ast(k,s) =
\begin{cases}
f_1^\ast(s), & k=1 \\
f_k^\ast(s) \prod_{j=1}^{k-1} \mathcal{H}_\ast(j,s), & k>1
\end{cases}, \quad f_k^\ast \in \{\sin, \cos\} \\
&X_{\text{particles}} = \mathcal{F}[X_0, \mathcal{R}, \{\delta_i\}], \quad
\mathcal{F}[X_0, \mathcal{R}, \{\delta_i\}] = \bigoplus_i \mathcal{R}_i(X_0(\lambda+\delta_i)) \\
&[1em]

&\text{Universal Indicatrix / Kinematics:} \\
&y = r R x_0, \quad u = \dot{r} R x_0 + r \Omega R x_0, \quad
e = r R, \quad g = e^\mathrm{T} \eta e, \quad D = \partial + \Omega, \quad D\psi = D \psi \\
&\psi_{\text{spin}} = \text{Sec}(S^3 \text{ to } S^2), \quad \oint d\theta = 2\pi n \quad [1em]

&\text{Mass / Topology:} \\
&m_{\text{eff}} = \mathcal{M}(Q, m_0, B, S(Q,n), \delta, n) \cdot (1 + \Delta_{\text{bridge}}), \quad
\mu = \frac{m_p}{m_e} = \frac{3}{2} B^5 \\
&\mathcal{M}(Q, m_0, B, S, \delta, n) = Q, m_0, B^{f(Q,\delta)} S(Q,n), \quad
f(Q,\delta) = \begin{cases} 4, & Q=1 \\ -1, & Q=3 \\ -5, & m_p/m_e \end{cases} \\
&S(Q,n) = B^{1-\delta}, \quad
Q = \sum_i W_i + L = 3A, \quad \Delta_{\text{bridge}} \approx 8.2 \times 10^{-5} \quad [1em]

&\text{Fundamental Constants / Cosmology:}
```

$$\begin{aligned} & \&B = \frac{3}{4}\pi \approx 0.23873241, \quad B_{\text{stable}} \approx 0.24177 \quad \\ & \&\tau_{\chi} \approx 1.45 \pm 0.20, \quad \ell_f = \left(\frac{2A}{T}\right)^{1/3} \quad \\ & \&L_{\text{UI}} = \frac{1}{2} \left(\partial_{\lambda} r\right)^2 + \frac{1}{2} r^2 \Omega_{\mu\nu} \\ & \Omega^{\mu\nu} = 0.5 \quad \\ & \&c_T = c \rightarrow c_1/c_2 = \text{fixed}, \quad Q \leq 3 \quad \\ & \&\frac{G}{c^4} \rightarrow 8\pi \ell_f^2, \quad S = \frac{A}{4} = n, \quad \nabla^2 f = -\frac{l(l+2)}{R^2} f \quad \\ & \Delta\phi \approx 0.246 \text{ rad}, \quad H_0 \approx 71.2 \text{ km/s/Mpc}, \quad \Lambda \rightarrow 0, \quad \tau_i + \tau_j + \tau_k \equiv 0 \pmod{3} \quad [1\text{em}] \end{aligned}$$

$$\begin{aligned} & \&\text{Master SAT Lagrangian:} \quad \\ & \&L_{\text{SAT}} = \sum_{i,o} \Big[\frac{1}{2} \mu_i \big| \frac{d\lambda}{dH} \big|^2 + \sum_j \kappa F_{\text{braid}} + \alpha \big(\frac{d\lambda}{dH} \cdot T \big)^2 + V_{\text{geom}}(\lambda) \Big] \quad \\ & \&L_u = \int \Big[\frac{1}{2} (\partial u)^2 + U(u) \Big] d^4x, \quad K = \kappa_0 + \kappa_2 \quad \\ & \&\Omega^{\mu_{\backslash} \nu} = (\partial_{\lambda} R^{\mu_{\backslash} \alpha}) R^{\alpha_{\backslash} \nu}, \quad \Omega_{\mu} = A_{\mu}^a T^a + B_{\mu}^i \tau^i + C_{\mu} Y, \quad F_{\mu\nu} = \partial_{\mu} \Omega_{\nu} - \partial_{\nu} \Omega_{\mu} + [\Omega_{\mu}, \Omega_{\nu}], \quad J_{\mu} = *J_{\mu\nu\rho} \end{aligned}$$